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MODELING RL10 THRUST INCREASE WITH DENSIFIED LH_2 AND LOX PROPELLANTS

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Abstract

Densified propellant technology has been demonstrated to improve performance capabilities of launch vehicles and employment of densified propellants is under investigation within several launch vehicle programs. Analysis of rocket engine performance using densified propellants is a key component of the research necessary for optimization of engine design and is required in order to advance the maturity of densified propellant technology.

A model of the RL10A-3-3A rocket engine utilizing the Rocket Engine Transient Simulator (ROCETS) mathematical modeling program has been modified to simulate densified propellant inlet conditions to determine the effects on overall system operating conditions. The nominal operating conditions of the RL10A-3-3A will be compared to operating conditions obtained at various levels of densified propellant inlet conditions. This system analysis will determine the effects of the colder inlet conditions on the overall system performance and will identify rocket engine system component interactions as influenced by the presence of densified propellants.

The research results presented last year at the 38th AIAA/ASME/SAE/ASEE Joint Propulsion Conference (Ref. 1) quantified some of the key benefits for densified propellants in engine systems including pressure drop reduction and decreased turbine/pump speeds, which results in increased engine life and reliability. This second paper on this research will present the effects of densified propellant on increased engine thrust. Flowing densified hydrogen and oxygen through the RL10 turbo-machinery operating at nominal rotational speeds can result in a nominal increase in thrust level of 12 percent. This provides an additional degree-of-freedom for aerospace vehicle designers to determine the optimal usage of densified propellants; increased thrust or decreased turbo-machinery speeds and pressure drops. In addition, the RL10 model is being used to examine a variety of densified oxygen and hydrogen propellant inlet conditions for a given set of required engine operating conditions such as thrust level, mixture ratio, and chamber pressure.

Introduction

The great benefits of using densified liquid oxygen (LOX) and liquid hydrogen (LH_2) propellants in aerospace vehicles have been well documented (Refs. 1-7). The recent and quick acceleration of mature cryogenic fluid liquefaction and densification technology now provides the capability for aerospace vehicles to operate at a variety of densified propellants temperatures enabling “virtual-storable” cryogenic propellants (Ref. 8). The performance of a standard RL10 engine with densified propellants represents a great unknown for the engine community in the effort to begin operating aerospace vehicles using densified propellants. The previous RL10 study focused on the engine system benefits at nominal operating conditions such as the reduction of pressure drop, and turbopump speed (Ref. 1).

This study investigates engine thrust increase with densified propellants at the nominal pump speed of 32000 rpm. Propellant inlet temperatures are varied from 150-100 ^{0}R for LOX, and 38-26 ^{0}R for LH_2. The existing RL10-3-3A engine critical components such as thrust control valve (TCV) and mixture ratio control valve (MRCV) at nominal operating conditions are studied and compared to the model predictions. The ability to operate the existing RL10 engine in a certain range of densified propellant temperatures without requiring hardware modification is investigated. All the results of the study are presented and discussed. Background information on densified propellants and the RL10 rocket engine are presented in Reference 1.

RL10A-3-3A Model

Model Computation Algorithm

The RL10 model is a lumped parameter model based on the full expander thermodynamic cycle. The model computation consists of empirical models and experimental map data provided by Pratt & Whitney, the RL10 engine manufacturer (Refs. 9-12). The RL10 model consists of fixed inputs, variable inputs, and targets. The model uses the shooting computational method where the variable inputs are varied until targets are obtained for given fixed inputs. In this study, propellant temperatures are fixed inputs. Engine thrust, combustion chamber pressure, and O/F mixture ratio are targets. TCV and MRCV are variable inputs. Figure 1 shows the diagram of the RL10 computational algorithm.

The physical dimensions and geometries of engine components are modeled as fixed parameters in the model. For a given fixed input propellant temperature, to obtain setting targets such as engine thrust and O/F mixture ratio, fuel and LOX mass flow rates are required. The model seeks the targeted pump speeds base on mass flow rates. If the LOX and fuel pump speeds are obtained, but the setting targets engine thrust and O/F mixture ratio are still not satisfied, the variable inputs TCV and MRCV change simultaneously to manipulate the LOX and fuel mass flows until target areas are achieved.

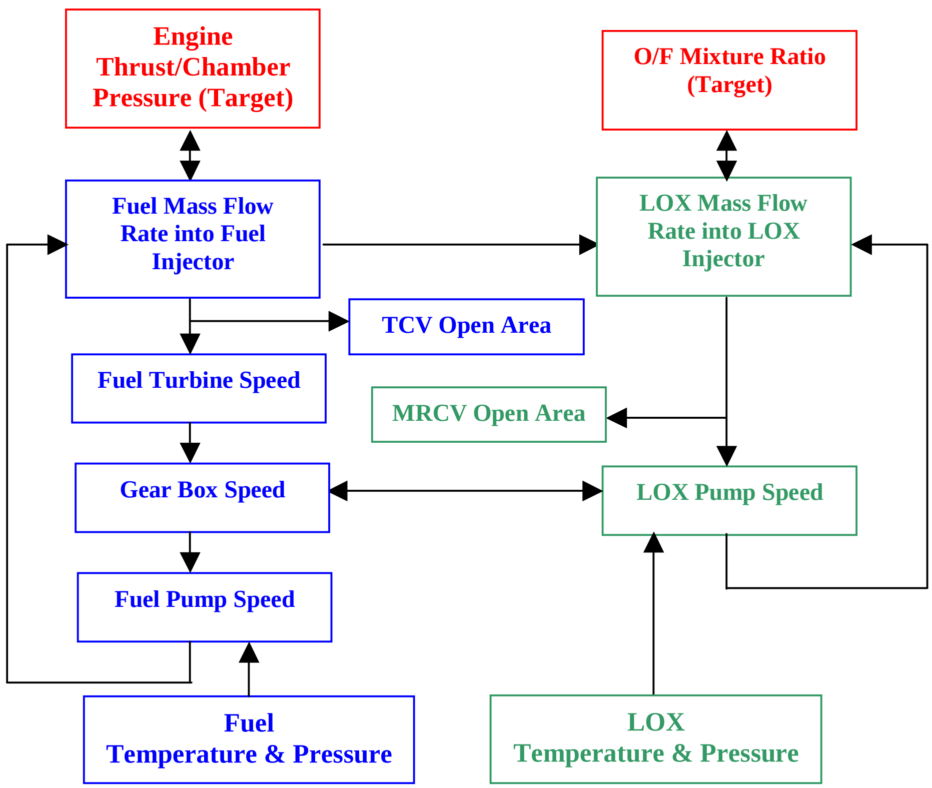

Figure 1 - The RL10 Engine Computational Algorithm Diagram

Model Verification

In the previous paper (Ref.1), the model prediction has been compared to the actual RL10 engine performance parameters at nominal operating condition, and the model shows an excellent agreement with test results (Refs. 1, 10, 11). Under nominal operating conditions, the model inputs are presented in Table 1.

In this paper, a more detailed study on engine component performance is conducted. The thrust control valve (TCV), and the O/F mixture ratio control valve (MRCV) open areas are investigated in the actual RL10 engine and compared to model predictions. Figures 2 and 3 show the excellent agreement between actual and predicted TCV and MRCV nominal open areas. Once again this confirms the accuracy of the RL10 model. Figure 3 also shows that it is impossible to operate the existing RL10 engine with LOX at 150 ^{0}R and LH_2 at 26 ^{0}R. The model predicts an open area larger than the maximum limit, thus indicating that the engine MRCV needs to be modified to operate at this propellant temperature combination.

Table 1- RL10A-3-3A Model Inputs at Nominal Operating Condition

Engine Thrust (lbf)	16588
Combustion Chamber Nominal Pressure (psia)	475.00
Inlet O/F Mixing Ratio	5.055
LOX Inlet Temperature (ºR)	174.70
LOX Inlet Pressure (psia)	35.30
Fuel Inlet Temperature (ºR)	38.60
Fuel Inlet Pressure (psia)	27.0

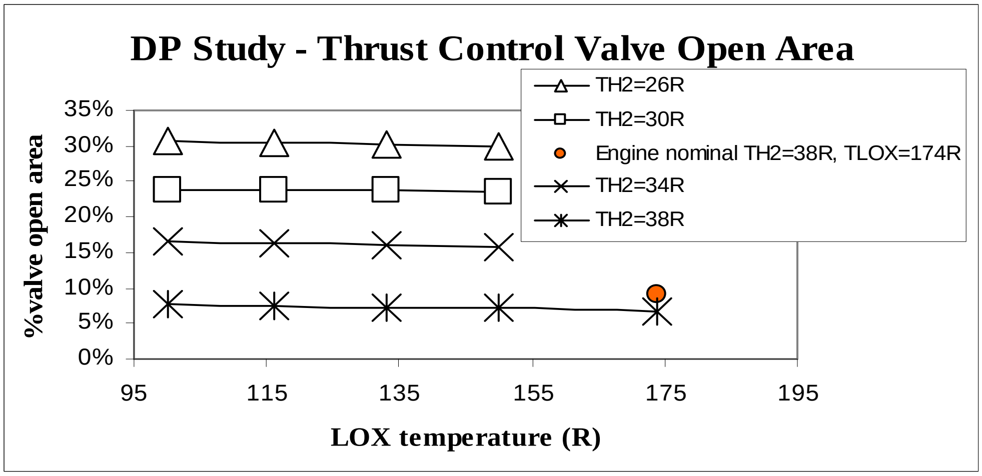

Figure 2 - Actual RL10 Engine and Model Prediction TCV % Open Area at Nominal Operating

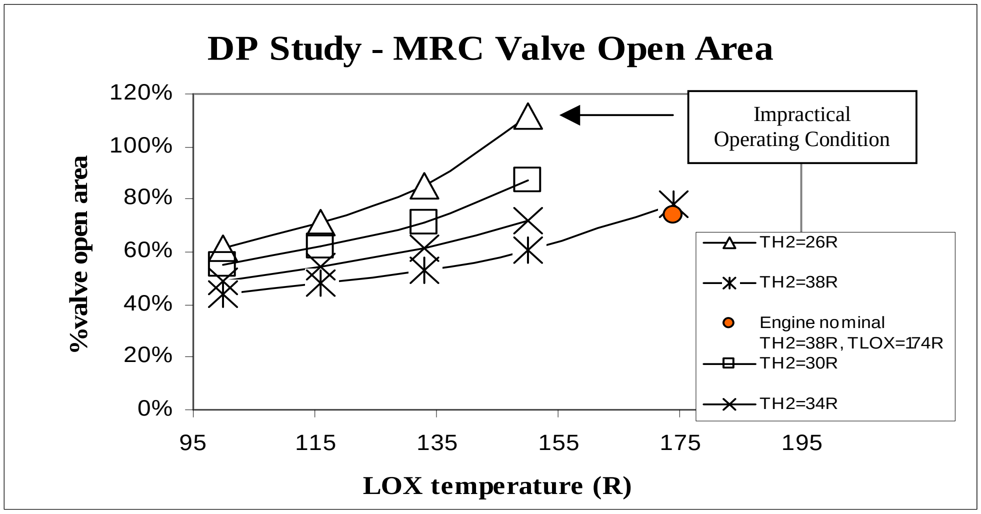

Figure 3 - Actual RL10 Engine and RL10 Model Prediction MRCV % Open Area at Nominal Operating Conditions

Model Input and Results Discussion

In this study, the LOX and fuel mass flow rates and combustion chamber pressure are adjusted manually to maintain a constant fuel pump speed of 32000 rpm at different inlet propellant temperatures. Table 2 shows the model input conditions for constant pump speed.

Figures 4 through 12 present performance predictions of various parameters and components of the RL10 engine as a function of propellant temperature at the nominal fuel pump speed of 32000 rpm. Figure 4 shows that the engine thrust increases approximately 12% compared to the normal boiling point propellant temperatures. This is due to the increase in mass flow rates of fuel and LOX (Figures 5 and 6) by approximately the same amount, 12%, due to the higher propellant density. The engine thrust increases significantly with lower fuel temperature, and increases slightly with LOX temperature. Therefore, in order to increase engine thrust, the liquid hydrogen fuel must be densified.

In Figures 7 and 8, the % valve open areas are normalized based on the maximum open areas of the actual engine valves. The thrust control valve (TCV) and O/F mixture ratio control valve (MRCV) open areas remain in the nominal operating range of the existing RL10 engine. Figures 9 and 10 show the TCV mass flow rate and the fuel bypass flow rate. As propellant temperature decreases, both flow rates increase. Figures 11 and 12 give engine turbine power, and turbine torque. Both increase with decreasing fuel temperature and decrease with decreasing LOX temperature resulting in only slightly higher values than nominal when both the LOX and fuel are densified.

Table 2 - RL10 Model Input for Constant Pumps Speed Study

Engine Fuel Pump Speed (rpm)	32000
O/F Mixture Ratio Control	5.055
LOX Inlet Pressure (psia)	42.35
LOX Inlet Temperature (^{0}R)	150-100
Fuel Inlet Pressure (psia)	27.68
Fuel Inlet Temperature (^{0}R)	38-26

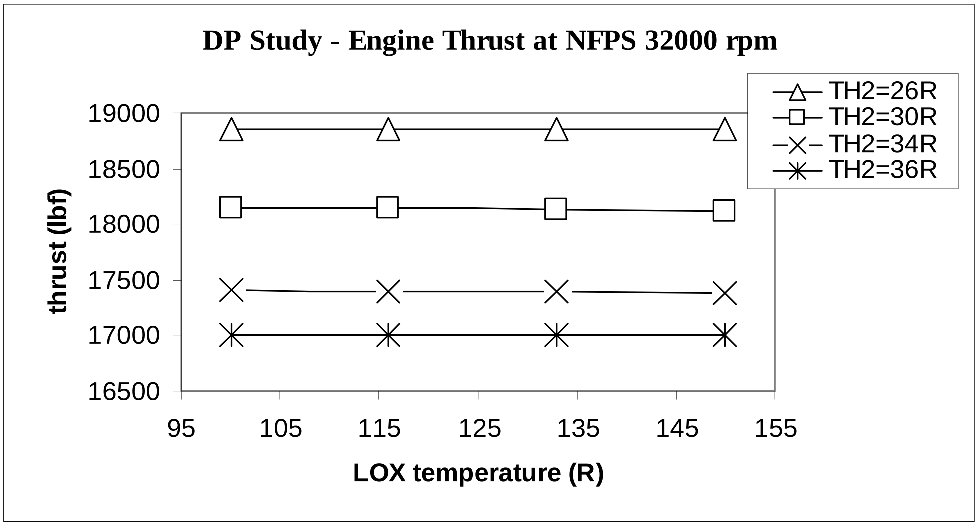

Figure 4 - Engine Thrust Increases with Densified Propellants at Nominal Pump Speed

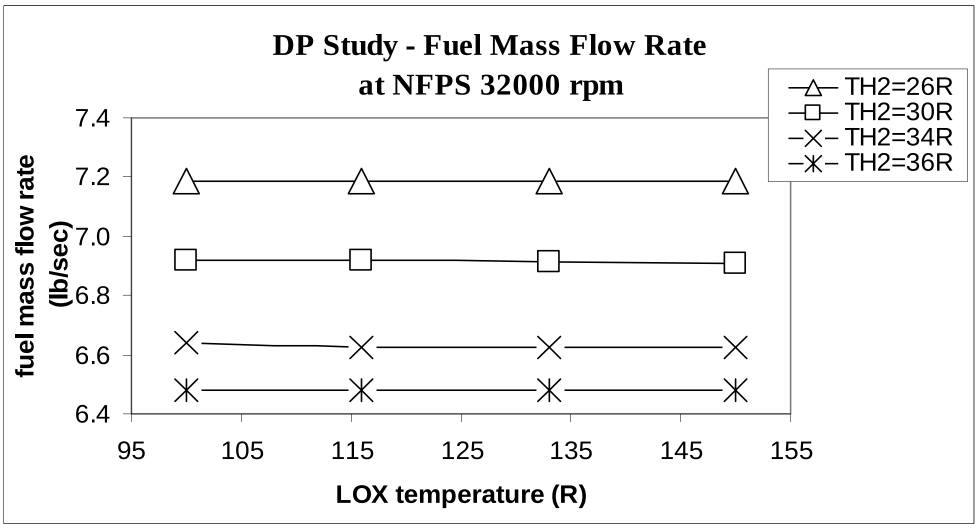

Figure 5 - Fuel Mass Flow Rate with Densified Propellants at Nominal Pump Speed

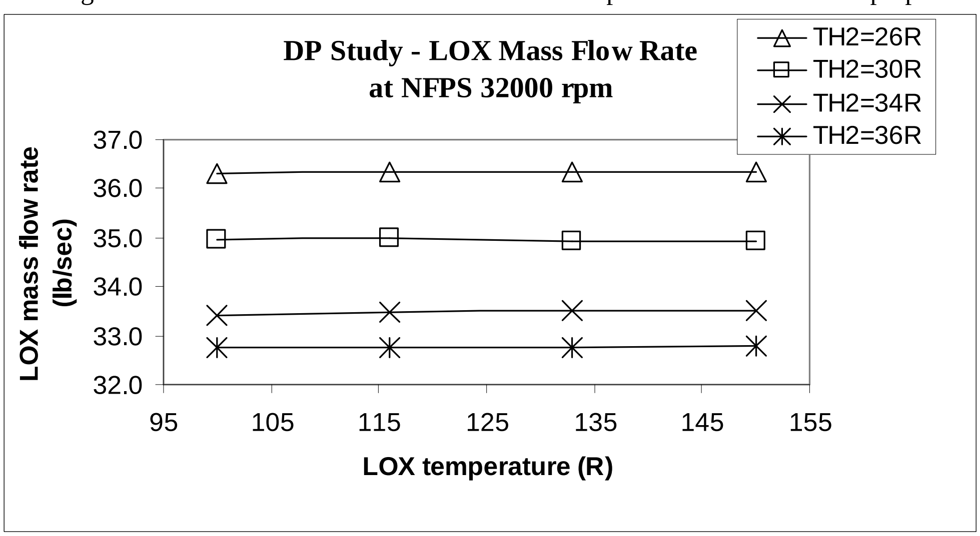

Figure 6 - LOX Mass Flow Rate with Densified Propellants at Nominal Pump Speed

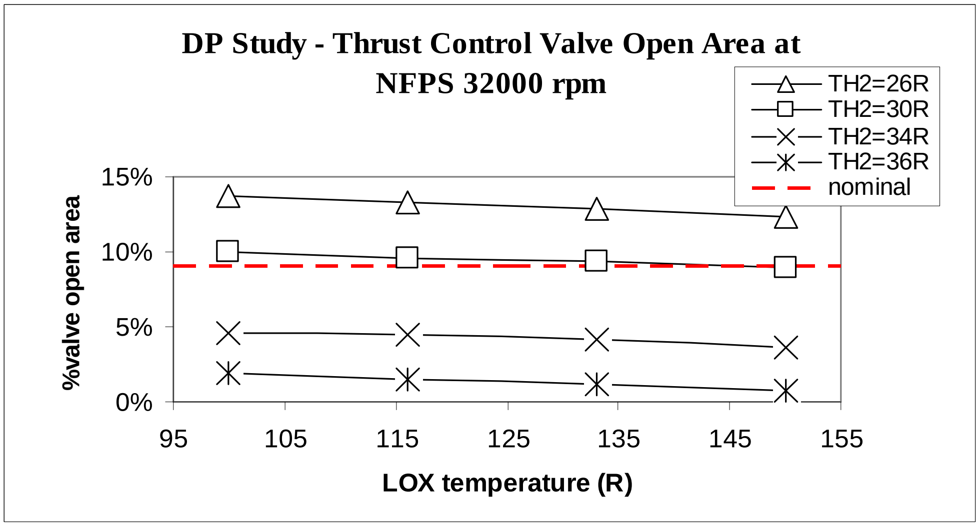

Figure 7 - TCV Open Area with Densified Propellants at Nominal Pump Speed

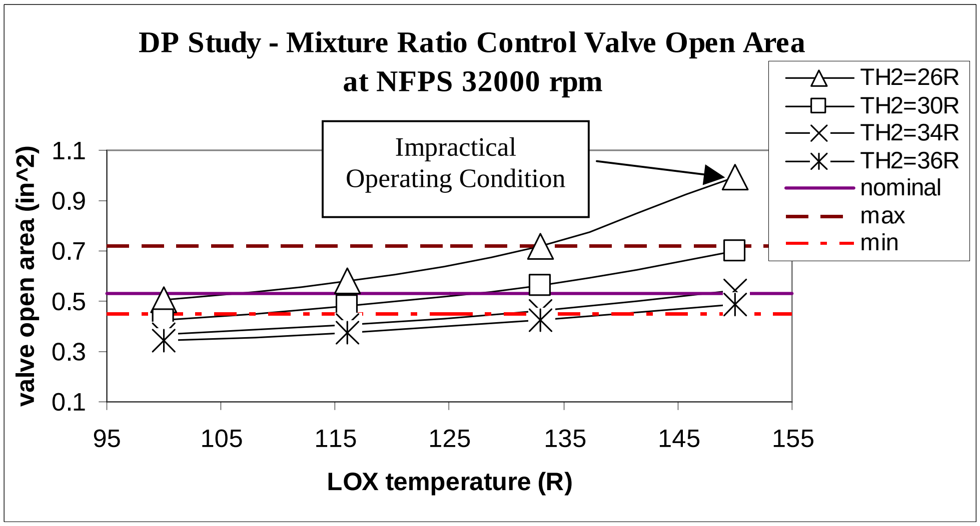

Figure 8 - MRC Valve Open Area with Densified Propellants at Nominal Pump Speed

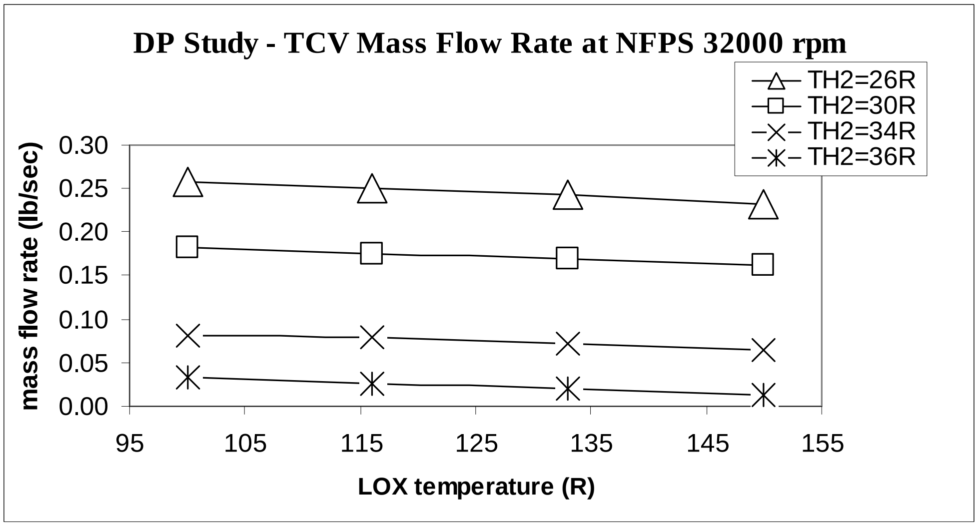

Figure 9 - TVC Mass Flow Rate with Densified Propellants at Nominal Pump Speed

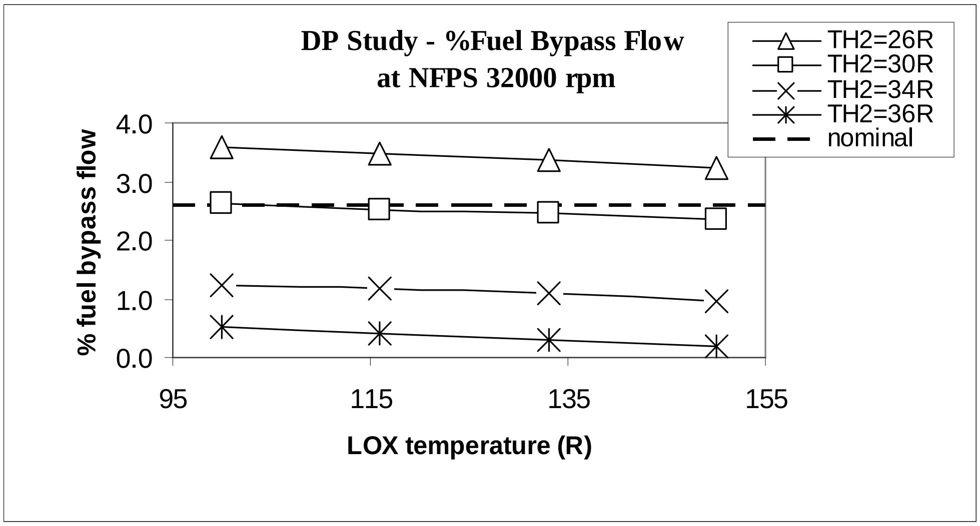

Figure 10 - % Fuel Bypass Flow with Densified Propellants at Nominal Pump Speed

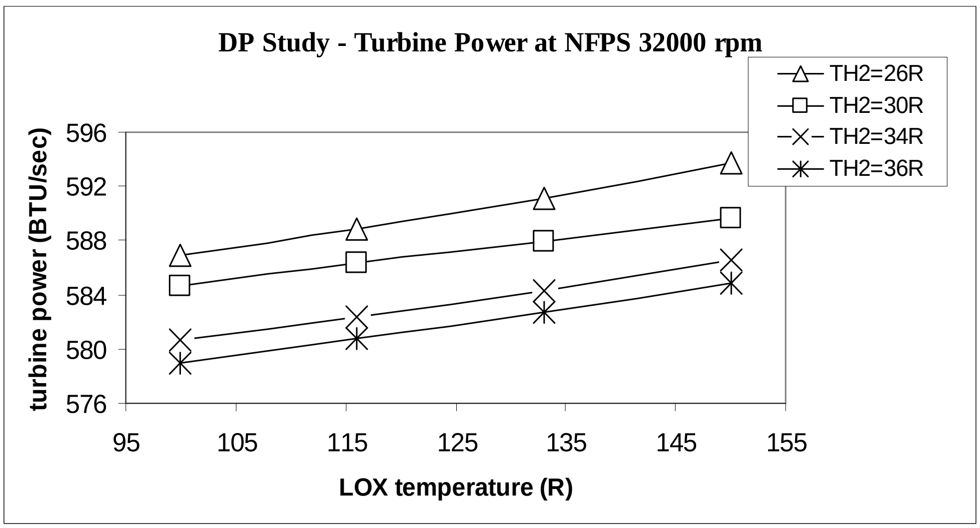

Figure 11 - Turbine Power with Densified Propellants at Nominal Pump Speed

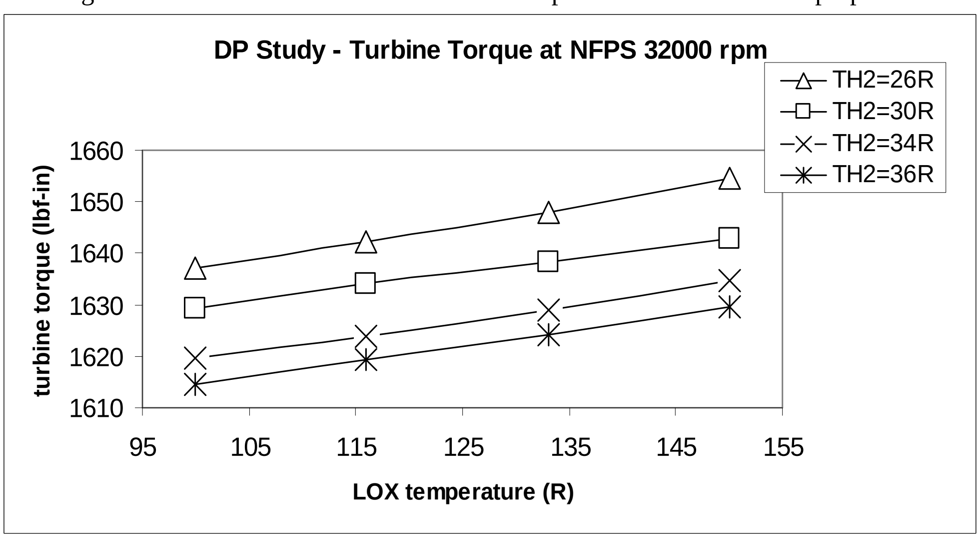

Figure 12 - Turbine Torque with Densified Propellants at Nominal Pump Speed

Conclusions

The RL10 model shows excellent agreement with the actual RL10 engine performance at the engine component level. The RL10 model indicates the possibility of increasing engine thrust by 12% using densified LOX and LH_2 propellants. Fuel temperature has a strong effect on the engine performance, and the fuel must be densified in order for the engine to increase thrust. The engine operation is better when LOX and fuel are densified relative to each other. It is not recommended to operate the engine with just one propellant densified and the other propellant near its normal boiling point.

The model predictions present a good case for operating an existing RL10 engine with densified propellants within a certain propellant temperature range without requiring engine component modifications. The thrust control valve and mixture ratio control valve operate within their nominal design margins with densified propellants. In addition, turbine power and torque are only slightly increased when both propellants are densified. The analysis shows for the first time that densified propellants have little or no impact on a RL10 engine power balance.

This study shows the remarkable benefits that can be obtained in engine as well as vehicle performance. The significant increase in engine thrust and propellant mass with an existing RL10 engine and vehicle can obtained with densified propellants. A propellant mass increase on the vehicle without increasing the size of the vehicle, translates into a significant increase in payload capability and mission efficiency.

Recommendations

Densified propellants are an enabling technology that allows launch vehicles to significantly increase payload capability. The effort to mature the final technology gap, by fully understanding rocket engine performance using densified propellants, is recognized to be very important. Therefore, it is recommended that more actual engine data be obtained and compared to model predictions. A further detail analyses on key engine components should be conducted to gain an even higher level of confidence. Numerical and experimental investigations into combustion stability and ignitability as well as full-scale RL10 engine testing with densified LOX and LH2 are recommended.

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